Compiled Solutions – Senior Division

1. 11-12 C1 Assembly Language

This program counts the number of times 2 is a divisor of 32 which is 5. 5

2. 12-13 C1 Assembly Language

This program adds the squares from 1 to 10, inclusive. Sum is 385. 385

3. 13-14 C1 Assembly Language

This program in BASIC is: 0

```
A=10:B=2:C=5:D=8
if A>B then A=A-B
if B*C<D then B=B+C
if A=D then D=0
print (A/B+C)*D
end
```

4. 14-15 C1 Assembly Language

This program determines if (21, 20, 23) is a Pythagorean triple. 0
It is not so a 0 is printed.

5. 15-16 C1 Assembly Language

This program finds the number of terms it takes for the sum of the 7
terms of the sequence to exceed 100. The sequence is 1, 2, 4, 8, 16, 32, 64, …
The sums are 1, 3, 7, 15, 31, 63, 127….